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SEMICONDUCTOR STRUCTURE HAVING DIELECTRIC LINER AND METHOD OF MANUFACTURING THE SAME

Abstract

The present application discloses a semiconductor structure and a method for fabricating the same. The semiconductor structure includes a substrate; a bit line structure disposed over the substrate; a plurality of capacitor contact structures disposed next to the bit line structure; a plurality of dielectric liners disposed in the capacitor contact structures, wherein each of the dielectric liners surrounds at least a portion of the corresponding capacitor contact structures; and a plurality of landing pad layers each disposed partially covering a top surface and a sidewall of the corresponding capacitor contact structure. The substrate includes an isolation layer, wherein a plurality of active areas are defined by the isolation layer, wherein a plurality of source regions and drain regions are disposed in the active areas.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a semiconductor structure and a method of manufacturing the semiconductor structure. Particularly, the present disclosure relates to a semiconductor structure having a dielectric liner and a method for forming the semiconductor structure.

DISCUSSION OF THE BACKGROUND

[0002] Semiconductor structures are used in a variety of electronic applications, such as personal computers, cellular phones, digital cameras, and other electronic equipment. The semiconductor structures are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductive layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements on the substrate. As the semiconductor industry has progressed into advanced technology process nodes in pursuit of greater device density, higher performance, and lower costs, challenges of precise control of lithography across a wafer have arisen, and product performance and production yield are accordingly limited.

[0003] This Discussion of the Background section is provided for background information only. The statements in this Discussion of the Background are not an admission that the subject matter disclosed in this Discussion of the Background section constitutes prior art to the present disclosure, and no part of this Discussion of the Background section may be used as an admission that any part of this application, including this Discussion of the Background section, constitutes prior art to the present disclosure.

SUMMARY

[0004] One aspect of the present disclosure provides a semiconductor structure. The semiconductor structure includes a substrate; a bit line structure disposed over the substrate; a plurality of capacitor contact structures disposed next to the bit line structure; a plurality of dielectric liners disposed in the capacitor contact structures, wherein each of the dielectric liners surrounds at least a portion of the corresponding capacitor contact structures; and a plurality of landing pad layers each disposed partially covering a top surface and a sidewall of a corresponding capacitor contact structure. The substrate includes an isolation layer, the isolation layer defines a plurality of active areas, and a plurality of source regions and drain regions are disposed in the active areas.

[0005] Another aspect of the present disclosure provides a method for fabricating a semiconductor structure. The method includes providing a substrate; forming a bit line structure on the substrate; forming a capacitor contact structure next to the bit line structure, wherein the capacitor contact structure protrudes from the substrate; recessing a top surface of the bit line structure; forming a landing pad layer partially covering a top surface of the capacitor contact structure and partially covering a sidewall of the capacitor contact structure.

[0006] Another aspect of the present disclosure provides a semiconductor structure. The semiconductor structure includes a polysilicon layer having a first surface and a second surface opposite to the first surface; a substrate disposed on the second surface of the polysilicon layer; a plurality of bit line structures disposed on the substrate; and a spacer structure disposed on lateral sidewalls of the bit line structure, wherein in a cross-sectional view perspective, the polysilicon layer includes a pair of dielectric liners disposed in the polysilicon layer and on lateral sidewalls of the spacer structure, and wherein the dielectric liners are separated from each other and pairs of the dielectric liners face toward each other.

[0007] Another aspect of the present disclosure provides a method for fabricating a semiconductor structure. The method includes receiving a substrate; forming a bit line structure on a top surface of the substrate; forming a spacer structure on the bit line structure, wherein the spacer structure includes a sacrificial layer sandwiched by a first dielectric layer and a second dielectric layer; forming a polysilicon layer over the top surface of the substrate and forming a pair of dielectric liners in the polysilicon layer, wherein the polysilicon layer has a first surface and a second surface opposite to the first surface; removing the sacrificial layer to form a gap between the first dielectric layer and the second dielectric layer; reducing a width of the gap; and forming a seal layer to seal the gap.

[0008] In conclusion, the application discloses a manufacturing method of a semiconductor structure and a semiconductor structure thereof. The presence of a dielectric liner of the semiconductor structure prevents a storage leakage from a bit line structure to a polysilicon layer, and correctness of data stored in the bit line structure can be protected.

[0009] The foregoing has outlined rather broadly the features and technical advantages of the present disclosure in order that the detailed description of the disclosure that follows may be better understood. Additional features and technical advantages of the disclosure are described hereinafter, and form the subject of the claims of the disclosure. It should be appreciated by those skilled in the art that the concepts and specific embodiments disclosed may be utilized as a basis for modifying or designing other structures, or processes, for carrying out the purposes of the present disclosure. It should also be realized by those skilled in the art that such equivalent constructions do not depart from the spirit or scope of the disclosure as set forth in the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] A more complete understanding of the present disclosure may be derived by referring to the detailed description and claims. The disclosure should also be understood to be coupled with the figures' reference numbers, which refer to similar elements throughout the description.

[0011] FIG. **1** is a schematic cross-sectional diagram of a semiconductor structure in accordance with some embodiments of the present disclosure.

[0012] FIG. **2** is a flow diagram illustrating a method for manufacturing a semiconductor structure in accordance with some embodiments of the present disclosure.

[0013] FIGS. **3** to **29** are schematic cross-sectional diagrams of intermediate stages in the formation of a semiconductor structure in accordance with some embodiments of the present disclosure.

[0014] FIG. **30** is a schematic cross-sectional diagram of a semiconductor structure in accordance with alternative embodiments of the present disclosure.

[0015] FIG. **31** is a flow diagram illustrating a method of fabricating a semiconductor structure in accordance with alternative embodiments of the present disclosure.

[0016] FIG. **32** is a schematic cross-sectional diagram of a semiconductor structure in accordance with alternative embodiments of the present disclosure.

[0017] FIG. **33** is a flow diagram illustrating a method of fabricating a semiconductor structure in accordance with alternative embodiments of the present disclosure.

[0018] FIGS. **34** to **36** are schematic cross-sectional diagrams illustrating intermediate stages for fabricating the semiconductor in FIG. **32** in accordance with alternative embodiments of the present disclosure.

DETAILED DESCRIPTION

[0019] Embodiments, or examples, of the disclosure illustrated in the drawings are now described using a specific language. It shall be understood that no limitation of the scope of the disclosure is hereby intended. Any alteration or modification of the described embodiments, and any further applications of principles described in this document, are to be considered as normally occurring to one of ordinary skill in the art to which the disclosure relates. Reference numerals may be repeated throughout the embodiments, but this does not necessarily mean that feature(s) of one embodiment apply to another embodiment, even if they share the same reference numeral.

[0020] It shall be understood that, although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers or sections, these elements, components, regions, layers or sections are not limited by these terms. Rather, these terms are merely used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the present inventive concept.

[0021] The terminology used herein is to describe particular example embodiments only and is not intended to be limiting to the present inventive concept. As used herein, the singular forms "a," "an" and "the" are intended to include the plural forms as well, unless the context indicates otherwise. It should be understood that the terms "comprises" and "comprising," when used in this specification, point out the presence of stated features, integers, steps, operations, elements, or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, or groups thereof.

[0022] As the semiconductor industry has progressed into advanced technology process nodes in pursuit of greater device density, it has reached an advanced precision of photolithography. In order to further reduce device sizes, dimensions of elements and distances between different elements have to be proportionally reduced. However, with the reductions in the dimensions of the elements and the distances between different elements, challenges of precise control of the dimensions and the distances have arisen. For instance, a landing pad can be disconnected by a sharp corner of a bit line structure after an etching operation.

[0023] FIG. **1** is a schematic cross-sectional diagram of a semiconductor structure in accordance with some embodiments of the present disclosure. The semiconductor structure may include a substrate **11**, a first bit line structure **21** disposed over the substrate **11**, a second bit line structure **22** disposed over the substrate **11**, a polysilicon layer **41** disposed over the substrate **11** and surrounded by the first bit line structure **21** and the second bit line structure **22**, a dielectric liner **42** surrounding at least a portion **417** of the polysilicon layer **41**, and a landing pad **51** disposed over the polysilicon layer **41**, the dielectric liner **42** and the second bit line structure **22**.

[0024] In some embodiments, the substrate **11** may have a multilayer structure, or the substrate **11** may include a multilayer compound semiconductor structure. In some embodiments, the substrate **11** includes semiconductor structures, electrical components, electrical elements, or a combination thereof. In some embodiments, the substrate **11** includes transistors or functional units of transistors. In some embodiments, the substrate **11** includes active components, passive components, and/or conductive elements. The active components may include a memory die (e.g., a dynamic random-access memory (DRAM) die, a static random-access memory (SRAM) die, etc.), a power management die (e.g., a power management integrated circuit (PMIC) die), a logic die (e.g., a system-on-a-chip (SoC), a central processing unit (CPU), a graphics processing unit (GPU), an application processor (AP), a microcontroller, etc.), a radio frequency (RF) die, a sensor die, a micro-electro-mechanical-system (MEMS) die, a signal processing die (e.g., a digital signal processing (DSP) die), a front-end die (e.g., an analog front-end (AFE) die) or other active components. The passive components may include a capacitor, a resistor, an inductor, a fuse or other passive components. The conductive elements may include metal lines, metal islands, conductive vias, contacts or other conductive elements.

[0025] The active components, passive components, and/or conductive elements as mentioned above can be formed in and/or over a semiconductor substrate. The semiconductor substrate may be a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like. The semiconductor substrate can include an elementary semiconductor including silicon or germanium in a single crystal form, a polycrystalline form, or an amorphous form; a compound semiconductor material including at least one of silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and indium antimonide; an alloy semiconductor material including at least one of SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and GaInAsP; any other suitable materials; or a combination thereof. In some embodiments, the alloy semiconductor substrate may be a SiGe alloy with a gradient SiGe feature in which Si and Ge compositions change from one ratio at one location to another ratio at another location of the gradient SiGe feature. In some embodiments, the SiGe alloy is formed over a silicon substrate. In some embodiments, a SiGe alloy can be mechanically strained by another material in contact with the SiGe alloy.

[0026] The first bit line structure **21** includes a first conductive layer **211**, a second conductive layer **212** disposed over the first conductive layer **211**, and a first dielectric layer **213** disposed over the second conductive layer **212**. In some embodiments, a height of the first dielectric layer **213** is greater than a height H**3** of the second conductive layer **212**. In some embodiments, the height of the second conductive layer **212** is greater than a height H**1** of the first conductive layer **211**. In some embodiments, the first conductive layer **211** includes aluminum (Al), copper (Cu), tungsten (W), titanium (Ti), tantalum (Ta), titanium aluminum alloy (TiAl), titanium aluminum nitride (TiAlN), tantalum carbide (TaC), tantalum carbon nitride (TaCN), tantalum silicon nitride (TaSiN), manganese (Mn), zirconium (Zr), titanium nitride (TiN), tungsten nitride (WN), tantalum nitride (TaN), ruthenium (Ru), titanium silicon nitride (TiSiN), or other suitable materials. In some embodiments, the first conductive layer **211** includes tungsten (W). In some embodiments, the second conductive layer **212** includes a material different from a material of the first conductive layer **211**. In some embodiments, the first dielectric layer **213** includes silicon nitride, metallic nitride, or a combination thereof.

[0027] The second bit line structure **22** includes a second dielectric layer **221**, a third conductive layer **222** disposed over the second dielectric layer **221**, and a third dielectric layer **223** disposed over the third conductive layer **222**. In some embodiments, a height of the third dielectric layer **223** is greater than a height H**4** of the third conductive layer **222**. In some embodiments, the height H**4** of the third conductive layer **222** is greater than a height H**2** of the second dielectric layer **221**. In some embodiments, the second dielectric layer **221** includes silicon nitride, metallic nitride, or a combination thereof. In some embodiments, the third dielectric layer **223** includes a nitride material same as a nitride material of the second dielectric layer **221**. In some embodiments, the third conductive layer **222** includes aluminum (Al), copper (Cu), tungsten (W), titanium (Ti), tantalum (Ta), titanium aluminum alloy (TiAl), titanium aluminum nitride (TiAIN), tantalum carbide (TaC), tantalum carbon nitride (TaCN), tantalum silicon nitride (TaSiN), manganese (Mn), zirconium (Zr), titanium nitride (TiN), tungsten nitride (WN), tantalum nitride (TaN), ruthenium (Ru), titanium silicon nitride (TiSiN), or other suitable materials. In some embodiments, the third conductive layer **222** includes tungsten (W).

[0028] In some embodiments, the first dielectric layer **213**, the second dielectric layer **221** and the third dielectric layer **223** include a same material. In some embodiments, the second conductive layer **212** and the third conductive layer **222** include a same material.

[0029] In some embodiments, the height H**1** of the first conductive layer **211** of the first bit line structure **21** is substantially equal to the height H**2** of the second dielectric layer **221** of the second bit line structure **22**. In some embodiments, the height H**3** of the second conductive layer **212** of the first bit line structure **21** is substantially equal to the height H**4** of the third conductive layer **222** of the second bit line structure **22**.

[0030] The semiconductor structure may further include a first spacer **31** surrounding the first bit line structure **21**, and a second spacer **32** surrounding the second bit line structure **22**. In some embodiments, detailed structures and configurations of the first spacer **31** and the second spacer **32** are substantially identical. For brevity, only the first spacer **31** is described in the following description, and a detailed description of the second spacer **32** is omitted herein. However, such omission is not intended to limit the present disclosure.

[0031] In some embodiments, the first spacer **31** may be a multi-layer **39** structure. In some embodiments, the first spacer **31** includes a first nitride layer **33**, an oxide layer **34** and a second nitride layer **35**. In some embodiments, the oxide layer **34** is sandwiched between the first nitride layer **33** and the second nitride layer **35**. In some embodiments, a thickness of the first nitride layer **33** is substantially equal to a thickness of the second nitride layer **35**. In some embodiments, a thickness of the oxide layer **34** is less than that of the first nitride layer **33** or the second nitride layer **35**. In some embodiments, the first nitride layer **33** and the second nitride layer **35** include a same nitride material. In some embodiments, the oxide layer **34** includes silicon oxide. In some embodiments, the first nitride layer **33** or the second nitride layer **35** includes silicon nitride.

[0032] In some embodiments, the first spacer **31** is disposed along a sidewall **217** of the first bit line structure **21**. In some embodiments, the second spacer **32** is disposed along a sidewall **227** of the second bit line structure **22**. In some embodiments, the first spacer **31** and the second spacer **32** surround the polysilicon layer **41** and the dielectric liner **42**.

[0033] The polysilicon layer **41** is disposed over the substrate **11** and surrounded by the first bit line structure **21** and the second bit line structure **22**. In some embodiments, the polysilicon layer **41** is disposed between the first bit line structure **21** and the second bit line structure **22** adjacent to the first bit line structure **21**. In some embodiments, a top surface **453** of the polysilicon layer **41** is higher than the second conductive layer **212** and higher than the third conductive layer **222**. In some embodiments, the top surface **453** of the polysilicon layer **41** is lower than a top surface **261** of the first bit line structure **21** and lower than a top surface **262** of the second bit line structure **22**. The polysilicon layer **41** can function as a contact for forming electrical connections to other electrical components, devices or elements in the substrate **11**. In some embodiments, the polysilicon layer **41** may include multiple portions (i.e., the polysilicon layer **41** between the first bit line structure **21** and the second bit line structure **22** can be one of the multiple portions) electrically isolated from one another, and different portions of the polysilicon layer **41** may electrically connect to different electrical components, devices or elements in the substrate **11**.

[0034] In some embodiments, the first spacer **31** and the second spacer **32** protrude from the top surface **453** of the polysilicon layer **41**. In some embodiments, a top surface **311** of the first spacer **31** and a top surface **321** of the second spacer **32** are higher than the top surface **453** of the polysilicon layer **41**.

[0035] As mentioned above, the dielectric liner **42** surrounds the portion **417** of the polysilicon layer **41**. In some embodiments, the dielectric liner **42** is disposed along a sidewall **415** of the polysilicon layer **41**. In some embodiments, a thickness of the dielectric liner **42** is substantially less than half of a width of the polysilicon layer **41**.

[0036] In some embodiments, a top surface **421** of the dielectric liner **42** is higher than a top surface **215** of the second conductive layer **212**, and a bottom surface **422** of the dielectric liner **42** is lower than a bottom surface **216** of the second conductive layer **212**. In other words, a height of the dielectric liner **42** is substantially greater than the height H**3** of the second conductive layer **212** of the first bit line structure **21**.

[0037] In some embodiments, the top surface **421** of the dielectric liner **42** is higher than a top surface **225** of the third conductive layer **222**, and the bottom surface **422** of the dielectric liner **42** is lower than a bottom surface **226** of the third conductive layer **222**. In other words, the height of the dielectric liner **42** is substantially greater than the height H**4** of the third conductive layer **222** of the second bit line structure **22**.

[0038] One or more of the landing pads **51** may be disposed over the polysilicon layer **41**, the dielectric liner **42** and the second bit line structure **22**. In some embodiments, the landing pad **51** includes one or more metallic materials, such as aluminum (Al), copper (Cu), tungsten (W), titanium (Ti), tantalum (Ta), titanium aluminum alloy (TiAl), titanium aluminum nitride (TiAlN), tantalum carbide (TaC), tantalum carbon nitride (TaCN), tantalum silicon nitride (TaSiN), manganese (Mn), zirconium (Zr), titanium nitride (TiN), tungsten nitride (WN), tantalum nitride (TaN), ruthenium (Ru), titanium silicon nitride (TiSiN), other suitable materials, or a combination thereof. In some embodiments, each of the landing pads **51** is disposed over a corresponding portion of the polysilicon layer **41**. In some embodiments, the landing pads **51** are electrically isolated from one another.

[0039] In some embodiments, the landing pad **51** includes a neck portion **511** aligned with the top surface **262** of the second bit line structure **22**. In some embodiments, the neck portion **511** is a portion of the landing pad **51** with a smallest width. In some embodiments, the top surface **421** of the dielectric liner **42** is lower than the neck portion **511** of the landing pad **51**.

[0040] FIG. **2** is a flow diagram illustrating a method S**1** for manufacturing a semiconductor structure in accordance with some embodiments of the present disclosure. The method S**1** includes a number of operations (S**101**, S**102**, S**103**, S**104**, S**105**, S**106**, S**107**, S**108**, S**109**, S**110** and S**111**) and the description and illustration are not deemed as a limitation to the sequence of the operations. In operation S**101**, a substrate is provided. In operation S**102**, a first conductive layer is disposed over the substrate. In operation S**103**, portions of the first conductive layer are removed. In operation S**104**, a second dielectric layer is disposed adjacent to the first conductive layer. In operation S**105**, a second conductive layer is disposed over the first conductive layer and the second dielectric layer, and a first dielectric layer is disposed over the second conductive layer. In operation S**106**, portions of the first dielectric layer, the second conductive layer and the second dielectric layer are removed, to form a first bit line structure comprising the first conductive layer, the second conductive layer and the first dielectric layer, to form a second bit line structure comprising the second dielectric layer, a third conductive layer and a third dielectric layer, and to form a recess between the first bit line structure and the second bit line structure. In operation S**107**, a first spacer surrounding the first bit line structure and a second spacer surrounding the second bit line structure are formed. In operation S**108**, a first polysilicon layer is formed within the recess and over the substrate. In operation S**109**, a dielectric liner is formed along a sidewall of the first spacer, a sidewall of the second spacer and over the first polysilicon layer. In operation S**110**, a second polysilicon layer is formed over the first polysilicon layer. In operation S**111**, portions of the dielectric liner are removed. It should be noted that the operations of the method S**1** may be rearranged or otherwise modified within the scope of the various aspects. Additional processes may be provided before, during, and after the method S**1**, and some other processes may be only briefly described herein. Thus, other implementations are possible within the scope of the various aspects described herein.

[0041] FIGS. **3** to **29** are schematic cross-sectional diagrams illustrating various fabrication stages constructed according to the method S**1** for manufacturing a semiconductor structure similar to that shown in FIG. **1** in accordance with some embodiments of the present disclosure. The stages shown in FIGS. **3** to **29** are also illustrated schematically in the process flow in FIG. **2**. In the subsequent discussion, the fabrication stages shown in FIGS. **3** to **29** are discussed with reference to the process steps in FIG. **2**.

[0042] Referring to FIG. **3**, FIG. **3** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. In operation S**101**, a substrate **11** is provided, received, or formed. In some embodiments, the substrate **11** may have a multilayer structure, or the substrate **11** may include a multilayer compound semiconductor structure. In some embodiments, the substrate **11** includes semiconductor structures, electrical components, electrical elements or a combination thereof. In some embodiments, the substrate **11** includes transistors or functional units of transistors. In some embodiments, the substrate **11** includes active components, passive components, and/or conductive elements. In some embodiments, the substrate **11** is similar to that shown in FIG. **1**. The substrate **11** can be formed following a conventional method for forming a semiconductor substrate.

[0043] Referring to FIG. **4**, FIG. **4** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. In some embodiments, the operation S**102** is performed after the operation S**101**. In operation S**102**, a first conductive layer **211** is disposed or formed over the substrate **11**.

[0044] Referring to FIGS. **5** to **8**, FIGS. **5** to **8** are schematic cross-sectional diagrams at a stage of the method S**1** in accordance with some embodiments of the present disclosure. In some embodiments, the operation S**103** is performed after the operation S**102** and includes multiple steps.

[0045] Referring to FIGS. **5** to **6**, FIGS. **5** to **6** are schematic cross-sectional diagrams at a stage of the method S**1** in accordance with some embodiments of the present disclosure. A patterned mask **102** is disposed over the first conductive layer **211** according to step S**103** in FIG. **2**. In some embodiments, the disposing of the patterned mask **102** includes disposing a photoresist **102′** over the first conductive layer **211** as shown in FIG. **5**, and then removing some portions of the photoresist **102′** to form the patterned mask **102** as shown in FIG. **6**.

[0046] In some embodiments, the photoresist **102′** is disposed by spin coating or any other suitable process. In some embodiments, some portions of the photoresist **102′** are removed by etching or any other suitable process. In some embodiments, at least a portion of the first conductive layer **211** is exposed through the patterned mask **102** after the formation of the patterned mask **102** as shown in FIG. **6**.

[0047] Referring to FIGS. **7** to **8**, FIGS. **7** to **8** are schematic cross-sectional diagrams at a stage of the method S**1** in accordance with some embodiments of the present disclosure. Portions of the first conductive layer **211** are removed. In some embodiments, the removal of the portions of the first conductive layer **211** includes etching or any other suitable process. In some embodiments, at least a portion of a surface **11***a* of the substrate **11** is exposed after the removal of the portions of the first conductive layer **211** as shown in FIG. **7**. In some embodiments as shown in FIG. **8**, after the removal of the portions of the first conductive layer **211**, the patterned mask **102** is removed by etching, stripping or any other suitable process.

[0048] Referring to FIG. **9**, FIG. **9** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. In operation S**104**, a second dielectric layer **221** is disposed or formed over the substrate **11** and adjacent to the first conductive layer **211**. In some embodiments, a top surface **211***a* of the first conductive layer **211** is substantially coplanar with a top surface **221***a* of the second dielectric layer **221**. In some embodiments, a height H**1** of the first conductive layer **211** is substantially equal to a height H**2** of the second dielectric layer **221**.

[0049] Referring to FIG. **10**, FIG. **10** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. In operation S**105**, a second conductive layer **212** is disposed over the first conductive layer **211** and the second dielectric layer **221**, and a first dielectric layer **213** is disposed over the second conductive layer **212**. In some embodiments, a thickness of the first dielectric layer **213** is greater than a thickness of the second conductive layer **212**. In some embodiments, the thickness of the second conductive layer **212** is greater than the height H**1** of the first conductive layer **211** and greater than the height H**2** of the second dielectric layer **221**.

[0050] In some embodiments, each of the second dielectric layer **221** and the first dielectric layer **213** includes one or more dielectric materials. In some embodiments, the dielectric material includes a polymeric material, an organic material, an inorganic material, a photoresist material, or a combination thereof. In some embodiments, the dielectric material includes one or more low-k dielectric materials having a dielectric constant (k value) less than 3.9. In some embodiments, the low-k dielectric material includes fluorine-doped silicon dioxide, organosilicate glass (OSG), carbon-doped oxide (CDO), porous silicon dioxide, spin-on organic polymeric dielectrics, spin-on silicon-based polymeric dielectrics, or a combination thereof. In some embodiments, the dielectric material includes one or more high-k dielectric materials having a dielectric constant (k value) greater than 3.9. The high-k dielectric material may include hafnium oxide (HfO.sub.2), zirconium oxide (ZrO.sub.2), lanthanum oxide (La.sub.2O.sub.3), yttrium oxide (Y.sub.2O.sub.3), aluminum oxide (Al.sub.2O.sub.3), titanium oxide (TiO.sub.2) or another applicable material. Other suitable materials are within the contemplated scope of this disclosure. In some embodiments, the dielectric material includes silicon oxide (SiO.sub.x), silicon nitride (Si.sub.xN.sub.y), silicon oxynitride (SiON), metallic nitride, or a combination thereof.

[0051] In some embodiments, the second dielectric layer **221** or the first dielectric layer **213** includes silicon nitride, metallic nitride, or a combination thereof. In some embodiments, the second dielectric layer **221** or the first dielectric layer **213** is formed by a blanket deposition. In some embodiments, the second dielectric layer **221** or the first dielectric layer **213** is formed by a chemical vapor deposition (CVD), a physical vapor deposition (PVD), an atomic layer deposition (ALD), a low-pressure chemical vapor deposition (LPCVD), a plasma-enhanced CVD (PECVD), or a combination thereof.

[0052] In some embodiments, the first conductive layer **211** or the second conductive layer **212** includes aluminum (Al), copper (Cu), tungsten (W), titanium (Ti), tantalum (Ta), titanium aluminum alloy (TiAl), titanium aluminum nitride (TiAlN), tantalum carbide (TaC), tantalum carbon nitride (TaCN), tantalum silicon nitride (TaSiN), manganese (Mn), zirconium (Zr), titanium nitride (TiN), tungsten nitride (WN), tantalum nitride (TaN), ruthenium (Ru), other applicable conductive materials, oxides of the above-mentioned metals, or a combination thereof. In some embodiments, the first conductive layer **211** or the second conductive layer **212** includes tungsten (W). In some embodiments, the second conductive layer **212** includes a material different from a material of the first conductive layer **211**. In some embodiments, the first conductive layer **211** or the second conductive layer **212** is formed by CVD, PVD, a sputtering operation, an electroplating operation, an electroless-plating operation, or a combination thereof.

[0053] Referring to FIGS. **11** to **14**, FIGS. **11** to **14** are schematic cross-sectional diagrams at a stage of the method S**1** in accordance with some embodiments of the present disclosure. In some embodiments, the operation S**106** is performed after the operation S**105** and includes multiple steps.

[0054] Referring to FIGS. **11** to **12**, FIGS. **11** to **12** are schematic cross-sectional diagrams at a stage of the method S**1** in accordance with some embodiments of the present disclosure. A patterned mask **104** is disposed over the first dielectric layer **213** according to step S**106** as shown in FIG. **12**. In some embodiments, the disposing of the patterned mask **104** includes disposing a photoresist **104′** over the first dielectric layer **213** as shown in FIG. **11**, and then removing some portions of the photoresist **104′** to form the patterned mask **104** as shown in FIG. **12**.

[0055] In some embodiments, the photoresist **104′** is disposed by spin coating or any other suitable process. In some embodiments, some portions of the photoresist **104′** are removed by etching or any other suitable process. In some embodiments, at least a portion of the first dielectric layer **213** is exposed through the patterned mask **104** after the formation of the patterned mask **104** as shown in FIG. **12**.

[0056] Referring to FIGS. **13** to **14**, FIGS. **13** to **14** are schematic cross-sectional diagrams at a stage of the method S**1** in accordance with some embodiments of the present disclosure. Portions of the first dielectric layer **213**, the second conductive layer **212** and the second dielectric layer **221** are removed.

[0057] In some embodiments, the removal of the portions of the first dielectric layer **213**, the second conductive layer **212** and the second dielectric layer **221** includes etching or any other suitable process.

[0058] In some embodiments, after the removal of the portions of the first dielectric layer **213**, the second conductive layer **212** and the second dielectric layer **221** as shown in FIG. **13**, a first bit line structure **21**, a second bit line structure **22** and a recess **61** between the first bit line structure **21** and the second bit line structure **22** are formed. The first bit line structure **21** includes the first conductive layer **211**, the second conductive layer **212** and the first dielectric layer **213**. The second bit line structure **22** includes the second dielectric layer **221**, the third conductive layer **222** and the third dielectric layer **223**. In some embodiments, a width W**1** of the first bit line structure **21** is substantially equal to a width W**2** of the second bit line structure **22**.

[0059] In some embodiments, as shown in FIG. **14**, after the removal of the portions of the first dielectric layer **213**, the second conductive layer **212** and the second dielectric layer **221**, the patterned masks **104** are removed by etching, stripping, or any other suitable process.
[0060] Referring to FIGS. **15** to **16**, FIGS. **15** to **16** are schematic cross-sectional diagrams at a stage of the method S**1** in accordance with some embodiments of the present disclosure. In some embodiments, operation S**107** is performed after the operation S**106** and includes multiple steps.
[0061] Referring to FIG. **15**, FIG. **15** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. After the formation of the first bit line structure **21** and the second bit line structure **22**, one or more conformal layers are formed over the first bit line structure **21**, the second bit line structure **22** and the substrate **11**. In some embodiments, each of the conformal layers includes a dielectric material, and two adjacent conformal layers may include different dielectric materials. In some embodiments, the dielectric material includes one or more low-k dielectric materials having a dielectric constant (k value) less than 3.9. In some embodiments, the low-k dielectric material includes fluorine-doped silicon dioxide, organosilicate glass (OSG), carbon-doped oxide (CDO), porous silicon dioxide, spin-on organic polymeric dielectrics, spin-on silicon-based polymeric dielectrics, or a combination thereof. In some embodiments, the dielectric material includes one or more high-k dielectric materials having a dielectric constant (k value) greater than 3.9. The high-k dielectric material may include hafnium oxide (HfO.sub.2), zirconium oxide (ZrO.sub.2), lanthanum oxide (La.sub.2O.sub.3), yttrium oxide (Y.sub.2O.sub.3), aluminum oxide (Al.sub.2O.sub.3), titanium oxide (TiO.sub.2) or another applicable material. Other suitable materials are within the contemplated scope of this disclosure.
[0062] As shown in FIG. **15**, in some embodiments, the multiple conformal layers include a first nitride layer **33**, a second nitride layer **35**, and an oxide layer **34** between the first nitride layer **33** and the second nitride layer **35**. In some embodiments, a profile of each of the first nitride layer **33**, the oxide layer **34** and the second nitride layer **35** is conformal to a profile of the first bit line structure **21**, the second bit line structure **22** and the substrate **11**. In some embodiments, the first nitride layer **33**, the oxide layer **34** and the second nitride layer **35** are individually formed by a chemical vapor deposition (CVD), a physical vapor deposition (PVD), an atomic layer deposition (ALD), a low-pressure chemical vapor deposition (LPCVD), a plasma-enhanced CVD (PECVD), or a combination thereof. In some embodiments, each of the first nitride layer **33**, the oxide layer **34** and the second nitride layer **35** is formed by a conformal deposition. In some embodiments, a thickness of the first nitride layer **33** is substantially equal to a thickness of the second nitride layer **35**. In some embodiments, a thickness of the oxide layer **34** is less than that of the first nitride layer **33** or the second nitride layer **35**. In some embodiments, the first nitride layer **33** and the second nitride layer **35** include a same nitride material. In some embodiments, the oxide layer **34** includes silicon oxide. In some embodiments, the first nitride layer **33** or the second nitride layer **35** includes silicon nitride.
[0063] Referring to FIG. **16**, FIG. **16** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. After the formation of the conformal layers (e.g., the first nitride layer **33**, the oxide layer **34** and the second nitride layer **35**), horizontal portions of the conformal layers are removed to form a first spacer **31** surrounding the first bit line structure **21** and a second spacer **32** surrounding the second bit line structure **22**. In some embodiments, the removal of the horizontal portions of the conformal layers includes a wet etching operation, a dry etching operation, or a combination thereof performed to form the first spacer **31** and the second spacer **32**. In some embodiments, the removal of the horizontal portions of the conformal layers includes a selective wet etching, a directional dry etching, an ion beam etching, a reactive ion etching, or a combination thereof.
[0064] In some embodiments, the horizontal portions of the second nitride layer **35**, the horizontal portions of the oxide layer **34**, and the horizontal portions of the first nitride layer **33** are

concurrently removed by one etching operation. In some embodiments, the horizontal portions of the second nitride layer **35**, the horizontal portions of the oxide layer **34**, and the horizontal portions of the first nitride layer **33** are individually removed by separate etching operations. The removal of the horizontal portions of the second nitride layer **35**, the oxide layer **34** and the first nitride layer **33** by multiple etching operations can be similar to the multiple etching operations of the formation of the first bit line structure **21** and the second bit line structure **22**, and repeated description is omitted herein. In some embodiments, the first spacer **31** surrounds sidewalls **217** of the first bit line structure **21**, and the second spacer surrounds sidewalls **227** of the second bit line structure **22**, as shown in FIG. **16**. In some embodiments, a top surface **261** of the first bit line structure **21** and a top surface **262** of the second bit line structure are exposed through the first spacer **31** and the second spacer **32**, respectively. In some embodiments, a height of the first spacer **31** is substantially equal to a height of the first bit line structure **21**, and a height of the second spacer **32** is substantially equal to a height of the second bit line structure **22**.

[0065] Referring to FIGS. **17** to **20**, FIGS. **17** to **20** are schematic cross-sectional diagrams at a stage of the method S**1** in accordance with some embodiments of the present disclosure. In some embodiments, the operation S**108** is performed after the operation S**107** and includes multiple steps.

[0066] Referring to FIG. **17**, FIG. **17** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. After the formation of the first spacer **31** and the second spacer **32**, in operation S**108**, a first polysilicon layer **41** is disposed within the recess **61** and over the substrate **11**. In some embodiments, the first polysilicon layer **41** is formed by a blanket deposition. In some embodiments, the blanket deposition includes a chemical vapor deposition (CVD), a physical vapor deposition (PVD), an atomic layer deposition (ALD), a low-pressure chemical vapor deposition (LPCVD), a plasma-enhanced CVD (PECVD), or a combination thereof. In some embodiments, the first polysilicon layer **41** covers the top surfaces **261** and **262** of the first bit line structure **21** and the second bit line structure **22**. In some embodiments, the first polysilicon layer **41** shown in FIG. **17** includes a height substantially greater than heights of the first bit line structure **21** and the second bit line structure **22**.

[0067] Referring to FIG. **18**, FIG. **18** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. After the deposition of the first polysilicon layer **41**, a sacrificial layer **43** is formed over the first polysilicon layer **41**. In some embodiments, the sacrificial layer **43** at least covers a top surface **419** of the first polysilicon layer **41**. It should be noted that FIG. **18** shows only a portion of the first polysilicon layer **41**, and the top surface **419** of the first polysilicon layer **41** may be a non-planar surface. The sacrificial layer **43** is configured to provide a planar surface for an etching or polishing operation to be performed in subsequent processing to provide a better result of planarization. In some embodiments, the sacrificial layer **43** has a top surface **431**, and the top surface **431** is planar. The sacrificial layer **43** is for a purpose of compensation of uneven portions of the top surface **419** of the first polysilicon layer **41**. In some embodiments, the sacrificial layer **43** includes a dielectric material, an anti-reflective coating material, an oxide-containing material, or other suitable materials. In some embodiments, the sacrificial layer **43** includes silicate glass, silicon oxide, silane oxide, or a combination thereof. In some embodiments, the sacrificial layer **43** includes borophosphosilicate glass (BPSG). In some embodiments, the sacrificial layer **43** includes a dielectric material different from that of the first dielectric layer **213** of the first bit line structure **21** or the third dielectric layer **223** of the second bit line structure **22**. In some embodiments, the sacrificial layer **43** includes a dielectric material different from that of the second nitride layer **35** of the first spacer **31** or the second spacer **32**. In some embodiments, the sacrificial layer **43** includes silicon.

[0068] Referring to FIG. **19**, FIG. **19** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. After the formation of the sacrificial layer **43**, a planarization is performed on the sacrificial layer **43** and the first polysilicon layer **41**. In some embodiments, the planarization includes ion beam etching, directional dry etching, reactive ion etching, solution wet etching, chemical mechanical polishing (CMP), or a combination thereof. In some embodiments, the planarization has a high selectivity to a material of the sacrificial layer **43**. In some embodiments, the planarization has a high selectivity to a material of the first polysilicon layer **41**. In some embodiments, the planarization has a low selectivity to a material of the first dielectric layer **213** and/or a material of the first spacer **31**. In some embodiments, the planarization stops upon an exposure of the top surface **261** of the first bit line structure **21** and the top surface **262** of the second bit line structure **22**. In some embodiments, a top surface **418** of the polysilicon layer **41** is substantially coplanar with the top surface **261** of the first bit line structure **21** and the top surface **262** of the second bit line structure **22**.

[0069] Referring to FIG. **20**, FIG. **20** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. After the planarization is performed on the sacrificial layer **43** and the first polysilicon layer **41**, portions of the first polysilicon layer are removed. In some embodiments, the removal of the portions of the first polysilicon layer **41** includes etching or any other suitable process. In some embodiments, a height H**5** of the first polysilicon layer **41** is obtained after the removal of the portions of the first polysilicon layer. In some embodiments, the height H**5** is less than the height H**1** of the first conductive layer **211** of the first bit line structure **21**, or is less than the height H**2** of the second dielectric layer **221** of the second bit line structure **22**. In some embodiments, the first bit line structure **21** or the second bit line structure **22** protrudes from and is exposed through the first polysilicon layer **41** after the removal of the portions of the first polysilicon layer **41**.

[0070] Referring to FIGS. **21** to **22**, FIGS. **21** to **22** are schematic cross-sectional diagrams at a stage of the method S**1** in accordance with some embodiments of the present disclosure. In some embodiments, the operation S**109** is performed after the operation S**108** and includes multiple steps.

[0071] Referring to FIG. **21**, FIG. **21** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. After the deposition of the first polysilicon layer **41**, a fourth dielectric layer **44** is formed over the first bit line structure **21**, the second bit line structure **22**, the first spacer **31**, the second spacer **32** and the first polysilicon layer **41**.

[0072] In some embodiments, the fourth dielectric layer **44** includes a dielectric material. In some embodiments, the dielectric material includes one or more low-k dielectric materials having a dielectric constant (k value) less than 3.9. In some embodiments, the low-k dielectric material includes fluorine-doped silicon dioxide, organosilicate glass (OSG), carbon-doped oxide (CDO), porous silicon dioxide, spin-on organic polymeric dielectrics, spin-on silicon-based polymeric dielectrics, or a combination thereof. In some embodiments, the dielectric material includes one or more high-k dielectric materials having a dielectric constant (k value) greater than 3.9. The high-k dielectric material may include hafnium oxide (HfO.sub.2), zirconium oxide (ZrO.sub.2), lanthanum oxide (La.sub.2O.sub.3), yttrium oxide (Y.sub.2O.sub.3), aluminum oxide (Al.sub.2O.sub.3), titanium oxide (TiO.sub.2) or another applicable material. Other suitable materials are within the contemplated scope of this disclosure. In some embodiments, the fourth dielectric layer **44** is formed by a conformal deposition.

[0073] Referring to FIG. **22**, FIG. **22** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. After the deposition of the fourth dielectric layer **44**, horizontal portions of the fourth dielectric layer **44** are removed to form a dielectric liner **42** along a sidewall **37** of the first spacer **31**, along a sidewall **38** of the second spacer **32**, and over the first polysilicon layer **41**. In some embodiments, the removal of the horizontal portions of the fourth dielectric layer **44** includes a wet etching operation, a dry etching operation, or a combination thereof performed to form the dielectric liner **42**. In some embodiments, the removal of the horizontal portions of the fourth dielectric layer **44** includes a selective wet etching, a directional dry etching, an ion beam etching, a reactive ion etching, or a combination thereof. In some embodiments, a bottom surface **422** of the dielectric liner **42** is substantially coplanar with a top surface **413** of the first polysilicon layer **41**.

[0074] Referring to FIGS. **23** to **24**, FIGS. **23** to **24** are schematic cross-sectional diagrams at a stage of the method S**1** in accordance with some embodiments of the present disclosure. In some embodiments, the operation S**110** is performed after the operation S**109** and includes multiple steps.

[0075] Referring to FIG. **23**, FIG. **23** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. After the formation of the dielectric liner **42**, a second polysilicon layer **45** is disposed over the first bit line structure **21**, the first spacer **31**, the dielectric liner **42**, the first polysilicon layer **41**, the second bit line structure **22** and the second spacer **32**. In some embodiments, the first polysilicon layer **41** and the second polysilicon layer **45** include a same material.

[0076] Referring to FIG. **24**, FIG. **24** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. After the deposition of the second polysilicon layer **45**, in operation S**110**, portions of the second polysilicon layer **45** are removed to form the second polysilicon layer **45**. In some embodiments, a height H**6** of the second polysilicon layer **45** is obtained after the removal of the portions of the second polysilicon layer **45**. In some embodiments, the first bit line structure **21** and the second bit line structure **22** protrude from and are exposed through the second polysilicon layer **45** after the removal of the portions of the second polysilicon layer **45**.

[0077] Referring to FIG. **25**, FIG. **25** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. In some embodiments, the operation S**111** is performed after the operation S**110**. After the formation of the second polysilicon layer **45**, portions of the dielectric liner **42** are removed. In some embodiments, a height H**7** of the dielectric liner **42** is obtained after the removal of the portions of the dielectric liner **42**. In some embodiments, the height H**7** of the dielectric liner **42** is substantially equal to the height H**6** of the second polysilicon layer. In some embodiments, a top surface **453** of the second polysilicon layer **45** is substantially coplanar with a top surface **421** of the dielectric liner **42**.

[0078] In some embodiments, the top surface **421** of the dielectric liner **42** is higher than a top surface **215** of the second conductive layer **212** of the first bit line structure **21**, and a bottom surface **422** of the dielectric liner **42** is lower than a bottom surface **216** of the second conductive layer **212** of the first bit line structure **21**. In other words, a height of the dielectric liner **42** is substantially greater than a height of the second conductive layer **212** of the first bit line structure **21**.

[0079] In some embodiments, the top surface **421** of the dielectric liner **42** is higher than a top surface **225** of the third conductive layer **222**, and the bottom surface **422** of the dielectric liner **42** is lower than a bottom surface **226** of the third conductive layer **222**. In other words, the height of the dielectric liner **42** is substantially greater than a height of the third conductive layer **222** of the second bit line structure **22**.

[0080] Referring to FIG. **26**, FIG. **26** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. After the removal of the portions of the dielectric liner **42**, the method S**1** further includes forming a metallic layer **52** covering the first bit line structure **21**, the second bit line structure **22**, the first spacer **31**, the second spacer **32**, the dielectric liner **42** and the second polysilicon layer **45**. In some embodiments, the metallic layer **52** includes aluminum (Al), copper (Cu), tungsten (W), titanium (Ti), tantalum (Ta), titanium aluminum alloy (TiAl), titanium aluminum nitride (TiAIN), tantalum carbide (TaC), tantalum carbon nitride (TaCN), tantalum silicon nitride (TaSiN), manganese (Mn), zirconium (Zr), titanium nitride (TiN), tungsten nitride (WN), tantalum nitride (TaN), ruthenium (Ru), titanium silicon nitride (TiSiN), other suitable materials, or a combination thereof. In some embodiments, the metallic layer **52** includes tungsten, copper, or a combination thereof. In some embodiments, the metallic layer **52** is formed by CVD, PVD, LPCVD, PECVD, a sputtering operation, electroplating, or a combination thereof. In some embodiments, the metallic layer **52** at least covers the top surface **261** of the first bit line structure **21** and the top surface **262** of the second bit line structure **22**. It should be noted that FIG. **26** shows only a portion of the metallic layer **52**, and a top surface **521** of the metallic layer **52** may be a non-planar surface.

[0081] Referring to FIG. **27**, FIG. **27** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. After the formation of the metallic layer **52**, the method S**1** may further include a planarization. In some embodiments, the planarization includes ion beam etching, directional dry etching, reactive ion etching, solution wet etching, CMP, or a combination thereof. In some embodiments, the planarization includes a polishing operation (e.g., a CMP operation). In some embodiments, a top surface **522** of the metallic layer **52** is formed after the planarization. In some embodiments, the top surface **522** is a planar surface disposed at an elevation lower than the top surface **521** shown in FIG. **26**.

[0082] Referring to FIG. **28**, FIG. **28** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. After the planarization, the method S**1** may further include removing portions of the metallic layer **52** to form a landing pad **51**. In some embodiments, a plurality of openings **53** are formed on the metallic layer **52** and thereby define a plurality of landing pads **51**. In some embodiments, an upper portion **313** of the first spacer **31** surrounding the first bit line structure **21** is partially removed during the removal of the portions of the metallic layer **52**. In some embodiments, the first spacer **31** and portions of the second spacer **32** are exposed through the plurality of openings **53**. In alternative embodiments, only portions of the metallic layer **52** are removed. In some embodiments, configurations of the first bit line structure **21**, the second bit line structure **22**, the first spacer **31** and the second spacer **32** remain the same before, during and after the removal of the portions of the metallic layer **52**.

[0083] In some embodiments, the landing pad **51** is disposed over the second polysilicon layer **45** and the second bit line structure **22**, and includes a neck portion **511** aligned with the top surface **262** of the second bit line structure **22**. In some embodiments, the top surface **421** of the dielectric liner **42** is below the neck portion **511** of the landing pad **51**.

[0084] Referring to FIG. **29**, FIG. **29** is a schematic cross-sectional diagram at a stage of the method S**1** in accordance with some embodiments of the present disclosure. After the formation of the landing pad **51**, the method S**1** may further include removing portions of the oxide layer **34** to form an air gap **36** surrounded by the first nitride layer **33**, the second nitride layer **35** and the oxide layer **34**. The air gap **36** is thereby formed in place of the removed portion of the oxide layer **34**. In some embodiments, the removal of the portions of the oxide layer **34** includes vapor etching, solution wet etching, or a combination thereof. In some embodiments, vapor-phase hydrogen fluoride (HF) is used to remove the portions of the oxide layer **34**. In some embodiments, a top surface **341** of the oxide layer **34** is lower than the bottom surface **422** of the dielectric liner **42**. A semiconductor structure similar to that shown in FIG. **1** is thereby formed.

[0085] Referring to FIG. **30**, FIG. **30** is a schematic cross-sectional diagram of a semiconductor structure **1**A in accordance with alternative embodiments of the present disclosure. The semiconductor structure **1**A may include a substrate **101**, a bit line structure **301** disposed on the substrate **101**, a capacitor contact structure **401** disposed next to the bit line structure **301**, and a landing pad layer **501** disposed covering a portion of a top surface **407**TS of the capacitor contact structure **401** and an upper portion of a sidewall **407**SW of the capacitor contact structure **401**. The semiconductor structure **1**A in FIG. **30** and the semiconductor structure in FIG. **1** are similar in many aspects, and thus descriptions of similar features will not be repeated herein. Main differences are described below.

[0086] The substrate **101** may include an organic semiconductor or a layered semiconductor such as silicon/silicon germanium, silicon-on-insulator or silicon germanium-on-insulator. When the substrate **101** is formed of silicon-on-insulator, the substrate **101** may include a top semiconductor layer and a bottom semiconductor layer formed of silicon, and a buried insulating layer which may separate the top semiconductor layer from the bottom semiconductor layer. The buried insulating layer may include, for example, a crystalline or non-crystalline oxide, nitride or any combination thereof.

[0087] In some embodiments, an isolation layer **103** may be formed in the substrate **101**, and active areas **105** may be defined by the isolation layer **103**. A top surface of the isolation layer **103** may be substantially coplanar with a top surface of the substrate **101**. The isolation layer **103** may be formed of, for example, an insulating material such as silicon oxide, silicon nitride, silicon oxynitride, silicon nitride oxide, or fluoride-doped silicate. It should be noted that, in the present disclosure, silicon oxynitride refers to a substance that contains silicon, nitrogen, and oxygen and in which a proportion of oxygen is greater than that of nitrogen. Silicon nitride oxide refers to a substance which contains silicon, oxygen, and nitrogen and in which a proportion of nitrogen is greater than that of oxygen. In some embodiments, source/drain regions **107-1** and **107-3** may be formed in upper portions of the active areas **105**. The source/drain regions **107-1**, **107-3** may be doped with a dopant such as phosphorus, arsenic, antimony, or boron.

[0088] The bit line structure **301** may include a bit line bottom conductive layer **303** disposed over the substrate **101**, a bit line middle conductive layer **305** disposed over the bit line bottom conductive layer **303**, and a bit line top conductive layer **307** disposed over the bit line middle conductive layer **305**. The bit line bottom conductive layer **303** may be formed of, for example, polycrystalline silicon, polycrystalline germanium, polycrystalline silicon germanium, titanium, tantalum, tungsten, copper, aluminum, tungsten silicide, cobalt silicide, or titanium silicide. The bit line middle conductive layer **305** may be formed of, for example, titanium nitride or tantalum nitride. The bit line top conductive layer **307** may be formed of, for example, tungsten, tantalum, titanium, copper, or aluminum. The bit line middle conductive layer **305** may reduce or possibly prevent a conductive material in the bit line top conductive layer **307** from diffusing toward the bit line bottom conductive layer **303**.

[0089] In some embodiments, the semiconductor structure **1**A further includes a bit line contact **311** formed in the substrate **101**, wherein the bit line structure **301** may be disposed on the bit line contact **311**. The bit line contacts **311** may be respectively correspondingly formed in the source regions **107-1**. A top surface of the bit line contact **311** may be substantially coplanar with the top surface of the substrate **101**. The bit line contact **311** may be formed of, for example, tungsten, cobalt, zirconium, tantalum, titanium, aluminum, ruthenium, copper, metal carbides (e.g., tantalum carbide, titanium carbide, tantalum magnesium carbide), metal nitrides (e.g., titanium nitride), transition metal aluminides, or a combination thereof. The bit line contact **311** may electrically couple to the source regions **107-1**.

[0090] In some embodiments, the semiconductor structure **1**A further includes bit line spacers **313** disposed protruding from the substrate **101** and attached to sidewalls of the bit line structure **301**. The bit line spacers **313** may be made of silicon oxide, silicon nitride, silicon carbon nitride, silicon nitride oxide, or silicon oxynitride. In some embodiments, some portions of a bottom surface **313**BS of the bit line spacer **313** may be substantially coplanar with a bottom surface **311**BS of the bit line contact **311**.

[0091] In some embodiments, the semiconductor structure **1**A further includes a bit line capping layer **309** over the substrate **101** and on the bit line structure **301**. A top surface **309**TS of the bit line capping layer **309** may be referred to as a top surface of the bit line structure **301**. The bit line capping layer **309** may be formed of, for example, silicon nitride, silicon nitride oxide, silicon oxynitride, boron nitride, silicon boron nitride, phosphorus boron nitride, or boron carbon silicon nitride.

[0092] The capacitor contact structure **401** may protrude from the substrate **101**. The capacitor contact structure **401** includes a capacitor contact bottom conductive layer **403** protruding from the substrate **101**, a capacitor contact middle conductive layer **405** disposed on the capacitor contact bottom conductive layer **403**, and a capacitor contact top conductive layer **407** disposed on the capacitor contact middle conductive layer **405**. The top surface **407**TS of the capacitor contact top conductive layer **407** may be referred to as the top surface of the capacitor contact structure **401**. The capacitor contact structure **401** may be electrically coupled to the drain region **107-3**. The capacitor contact bottom conductive layer **403** may be formed of, for example, polycrystalline silicon, polycrystalline germanium, or polycrystalline silicon germanium. In some embodiments, the capacitor contact bottom conductive layer **403** may be doped with a dopant such as phosphorus, arsenic, antimony, or boron. The capacitor contact middle conductive layer **405** may be formed of, for example, cobalt silicide, titanium silicide, nickel silicide, nickel platinum silicide, or tantalum silicide. A top surface of the capacitor contact middle conductive layer **405** may be at a vertical level lower than a vertical level of the top surface **309**TS of the bit line capping layer **309**.

[0093] In some embodiments, the semiconductor structure **1**A further includes a dielectric liner **402** surrounding at least a portion **412** of the capacitor contact bottom conductive layer **403** and a portion **432** of the capacitor contact middle conductive layer **405**. In some embodiments, the bit line spacer **313** surrounds the capacitor contact bottom conductive layer **403**, the capacitor contact middle conductive layer **405**, the capacitor contact top conductive layer **407** and the dielectric liner **402**. In some embodiments, the dielectric liner **402** is disposed along a sidewall **403**SW of the capacitor contact bottom conductive layer **403** and along a sidewall **405**SW of the capacitor contact middle conductive layer **405**.

[0094] The landing pad layer **501** may be formed partially covering the capacitor contact structure **401**. The landing pad layer **501** may cover a portion of the top surface **407**TS of the capacitor contact top conductive layer **407** and an upper portion of the sidewall **407**SW of the capacitor contact top conductive layer **407**. In other words, the landing pad layer **501** may partially cover the capacitor contact top conductive layer **407**. The landing pad layer **501** may be formed of tungsten, copper, or aluminum. A photolithography process and a subsequent etching process may be performed to form the landing pad layer **501**.

[0095] FIG. **31** is a flow diagram illustrating a method **10** for fabricating the semiconductor structure **1**A in accordance with alternative embodiments of the present disclosure. The method **10** may comprise processes S**11**, S**13**, S**15**, S**17** and S**19**, and descriptions of the processes and intermediate stages for fabricating the semiconductor **1**A are provided with reference to FIG. **30** for further understanding.

[0096] Referring to FIGS. **30** and **31**, in some embodiments, the process S**11** of the method **10** comprises: providing a substrate **101**; forming an isolation layer **103** in the substrate **101**; and defining a plurality of active areas **105** by the isolation layer **103**.

[0097] In some embodiments, the process S**13** of the method **10** comprises: forming bit line structures **301** on the substrate **101**.

[0098] In some embodiments, the formation of bit line structure **301** comprises: forming a bit line bottom conductive layer **303** over the substrate **101**; forming a bit line middle conductive layer **305** over the bit line bottom conductive layer **303**; forming a bit line top conductive layer **307** over the bit line middle conductive layer **305**; and forming a bit line capping layer **309** over the bit line top conductive layer **307**.

[0099] In some embodiments, the process S**15** of the method **10** comprises: forming a capacitor contact structure **401** next to the bit line structure **301** and protruding from the substrate **101**.

[0100] In some embodiments, the formation of the capacitor contact structure **401** comprises: forming a capacitor contact bottom conductive layer **403** protruding from the substrate **101**; forming a capacitor contact middle conductive layer **405** on the capacitor contact bottom conductive layer **403**; and forming a capacitor contact top conductive layer **407** on the capacitor contact middle conductive layer **405**.

[0101] In some embodiments, the process S**15** of the method **10** further comprises a process S**151**: forming a dielectric liner **402** along a sidewall **403**SW of the capacitor contact bottom conductive layer **403** and along a sidewall **405**SW of the capacitor contact middle conductive layer **405**, surrounding at least a portion **412** of the capacitor contact bottom conductive layer **403** and a portion **432** of the capacitor contact middle conductive layer **405**.
[0102] In some embodiments, the process S**17** of the method **10** comprises: recessing top surfaces **309**TS of the bit line structures **301**.
[0103] In some embodiments, the process S**19** of the method **10** comprises: forming landing pad layers **501** partially covering the capacitor contact structures **401**, wherein the landing pad layers **501** cover a portion of a top surface **407**TS of the capacitor contact structure **401** and an upper portion of a sidewall **407**SW of the capacitor contact structure **401**.
[0104] Referring to FIG. **32**, FIG. **32** is a schematic cross-sectional diagram of a semiconductor structure **1**B in accordance with alternative embodiments of the present disclosure. The semiconductor structure **1**B may include a polysilicon layer **14′**, a substrate **11′**, a bit line structure **12** and a spacer structure **13′**.
[0105] The polysilicon layer **14′** has a first surface F**14′** and a second surface B**14′** opposite to the first surface F**14′**. In some embodiments, the substrate **11′** is disposed on the second surface B**14′** of the polysilicon layer **14′**. In some embodiments, the bit line structure **12** is disposed on the substrate **11′**, penetrating through the polysilicon layer **14′** and protruding from the first surface F**14′** of the polysilicon layer **14′**.
[0106] The substrate **11′** and the bit line structure **12** are similar to the substrate **11** and the first bit line structure **21** (or the second bit line structure **22**) of the semiconductor structure illustrated in FIG. **1** and thus descriptions of the substrate **11′** and the bit line structure **12** will not be repeated herein.
[0107] The spacer structure **13′** is disposed on lateral sidewalls BS**1** and BS**2** of the bit line structure **12**. In some embodiments, the spacer structure **13′** may include a first dielectric layer **131′** and a second dielectric layer **133′**, wherein the second dielectric layer **133′** includes spacer portions **133***a* and **133***b* of a dielectric layer **133**, wherein the spacer portion **133***b* is disposed in the polysilicon layer **14′**, and the spacer portion **133***a* is disposed outside the polysilicon layer **14′**. In some embodiments, the spacer structure **13′** includes a gap **135** sandwiched by the first dielectric layer **131′** and the second dielectric layer **133′**. A thickness T**133***a′* of the spacer portion **133***a* of the dielectric layer **133** is less than a thickness T**133***b′* of the spacer portion **133***b* of the dielectric layer **133**. In some embodiments, a width T**135** of the gap **135** is in a range of 3 to 5 nanometers. In some embodiments, the thickness T**133***a′* of the spacer portion **133***a* is in a range of 4 to 8.5 nanometers. In some embodiments, the thickness T**133***b′* of the spacer portion **133***b* is in a range of 5.5 to 10 nanometers. In some embodiments, the thickness T**131′** of the first dielectric layer **131′** is in a range of 5.5 to 12 nanometers. In some embodiments, the thickness T**131′** of the first dielectric layer **131′** is substantially equal to the thickness T**133***b′* of the spacer portion **133***b*. In some embodiments, materials of the first dielectric layer **131′** and the second dielectric layer **133′** are the same. In some embodiments, the first dielectric layer **131′** and the second dielectric layer **133′** are made of nitride (e.g., silicon nitride). In some embodiments, the first dielectric layer **131′** and the second dielectric layer **133′** are made of oxide (e.g., silicon oxide).
[0108] In some embodiments, the semiconductor structure **1**B further includes a pair of dielectric liners **141′** disposed in the polysilicon layer **14′** and disposed on lateral sidewalls S**1**, S**2** of the spacer portion **133***b* of the dielectric layer **133**, wherein the dielectric liners **141′** are spaced apart from each other and face toward each other. In some embodiments, the pair of dielectric liners **141′** is disposed between a bit line structure **12** and an adjacent bit line structure **12**. In some embodiments, the dielectric liners **141′** are disposed to penetrate the polysilicon layer **14′**. In some embodiments, a top surface TS of the dielectric liner **141′** is substantially coplanar with the first surface F**14′** of the polysilicon layer **14′**. In some embodiments, a bottom surface BS of the

dielectric liner **141**′ is substantially coplanar with the second surface B**14**′ of the polysilicon layer **14**′. The dielectric liner **141**′ may be formed of materials same as those of the dielectric liner **42** of the semiconductor structure illustrated in FIG. **1**. A height H of the dielectric liner **141**′ may be equal to or less than a thickness T of the polysilicon layer **14**′.
[0109] In some embodiments, the semiconductor structure **1**B further includes a metal layer **15** disposed over the polysilicon layer **14**′, a landing pad **17**′ disposed over the metal layer **15**, and an adhesion layer **16**′ disposed between the metal layer **15** and the landing pad **17**′, wherein the adhesion layer **16**′ lines portions of top surfaces of the metal layer **15**, sidewalls of the spacer structure **13**′, top surfaces of the spacer structure **13**′, and portions of the bit line structure **12**.
[0110] In some embodiments, the semiconductor structure **1**B further includes a spacer layer **18** lining exposed portions of the metal layer **15**, the adhesion layer **16**′, the landing pad **17**′, the bit line structure **12**, and the spacer structure **13**′, and the semiconductor structure **1**B further includes a seal layer **19** disposed covering the spacer layer **18**.
[0111] FIG. **33** is a flow diagram illustrating a method M**10** for fabricating the semiconductor structure **1**B in accordance with alternative embodiments of the present disclosure. The method M**10** may comprise processes O**101**, O**103**, O**105**, O**107**, O**109**, O**111**, O**113** and O**115**, and descriptions of the processes and intermediate stages for fabricating the semiconductor are provided with reference to FIG. **32** and FIGS. **34** to **36** for further understanding.
[0112] Referring to FIGS. **33** and **34**, in some embodiments, the process O**101** of the method M**10** comprises receiving a substrate **11**′, wherein the substrate **11**′ is the same or similar to the substrate **11** of the semiconductor structure illustrated in FIG. **1**
[0113] In some embodiments, the process O**103** of the method M**10** comprises forming bit line structures **12** on a top surface F**11** of the substrate **11**′, wherein recessed portions R**1** are formed on the top surface F**11** of the substrate **11**′, adjacent to two lateral sides of one of the bit line structures **12**, and an adjacent bit line structure **12** is formed on a planar portion of the top surface F**11** of the substrate **11**′ with no recessed portions R**1** nearby.
[0114] In some embodiments, the process O**105** of the method M**10** comprises forming a spacer structure **13**′ on the bit line structure **12**, wherein the spacer structure **13**′ includes a sacrificial layer **132**′ sandwiched by a first dielectric layer **131**′ and a second dielectric layer **133**′.
[0115] In some embodiments, the process O**107** of the method M**10** comprises forming a polysilicon layer **14**′ over the top surface F**11** of the substrate **11**′, wherein the polysilicon layer **14**′ has a first surface F**14**′ and a second surface B**14**′ opposite to the first surface F**14**′; and forming a pair of dielectric liners **141**′ in the polysilicon layer **14**′ and on lateral sidewalls S**1**, S**2** of spacer portions **133***b* of the second dielectric layer **133**′, wherein the dielectric liners **141**′ are spaced apart from each other and face toward each other.
[0116] In some embodiments, the process O**109** of the method M**10** comprises forming a metal layer **15** over the polysilicon layer **14**′; forming an adhesion layer **16**′ lining portions of top surfaces of the metal layer **15**, sidewalls of the spacer structure **13**′, top surfaces of the spacer structure **13**′ and portions of the bit line structure **12**; and forming landing pads **17**′ over the metal layer **15** and on the adhesion layer **16**′.
[0117] Referring to FIGS. **33** and **35**, in some embodiments, the process O**111** of the method M**10** comprises removing the sacrificial layer **132**′ to form a gap **134** between the first dielectric layer **131**′ and the second dielectric layer **133**′, wherein the gap **134** has a width T**134**. Referring to FIGS. **33** and **36**, in some embodiments, the process O**113** of the method M**10** comprises reducing the width T**134** of the gap **134** by depositing a spacer layer **18** lining the intermediate structure shown in FIG. **35**. Thus, as shown in FIG. **36**, a width T**135** of the gap **135** is reduced. That is, the width T**135** of the gap **135** is less than the width T**134** of the gap **134**. It should be noted that, in some embodiments, the spacer layer **18** is made of the same material as the first dielectric layer **131**′ and/or the second dielectric layer **133**′. In some embodiments, there is no distinct interface between the spacer layer **18** and the first dielectric layer **131**′ if the spacer layer **18** and the first dielectric

layer **131**′ are made of the same material. In some embodiments, the spacer layer **18** may be formed by atomic layer deposition.

[0118] Referring to FIGS. **33** and **32**, in some embodiments, the process O**115** of the method M**10** comprises forming a seal layer **19** to seal the gap. In some embodiments, the seal layer **19** is a multi-layered structure. In some embodiments, the seal layer **19** includes a linear layer **191** and a planar layer **192**.

[0119] Therefore, the present disclosure provides a novel structure of a bit line structure and a method for manufacturing the same. The bit line structure of the present disclosure has a dielectric liner disposed along a sidewall of a polysilicon layer. The dielectric liner may prevent charges (e.g., charges stored in a capacitor configured on a landing pad) from leaking into the polysilicon layer. In addition, a neck portion of the landing pad may not need to be narrowed due to the deposition of the dielectric liner since a top surface of the dielectric liner is lower than the neck portion of the landing pad.

[0120] One aspect of the present disclosure provides a semiconductor structure. The semiconductor structure includes a substrate; a bit line structure disposed over the substrate; a plurality of capacitor contact structures disposed next to the bit line structure; a plurality of dielectric liners disposed in the capacitor contact structures, wherein each of the dielectric liners surrounds at least a portion of the corresponding capacitor contact structures; and a plurality of landing pad layers each disposed partially covering a top surface and a sidewall of the corresponding capacitor contact structure. The substrate includes an isolation layer, wherein the isolation layer defines a plurality of active areas, and a plurality of source regions and drain regions are disposed in the active areas.

[0121] Another aspect of the present disclosure provides a method for fabricating a semiconductor structure. The method includes providing a substrate; forming a bit line structure on the substrate; forming a capacitor contact structure next to the bit line structure, wherein the capacitor contact structure protrudes from the substrate; recessing a top surface of the bit line structure; and forming a landing pad layer partially covering a top surface of the capacitor contact structure and partially covering a sidewall of the capacitor contact structure.

[0122] Another aspect of the present disclosure provides a semiconductor structure. The semiconductor structure includes a polysilicon layer having a first surface and a second surface opposite to the first surface; a substrate disposed on the second surface of the polysilicon layer; a plurality of bit line structures disposed on the substrate; and a spacer structure disposed on lateral sidewalls of the bit line structure, wherein in a cross-sectional view perspective, the polysilicon layer includes a pair of dielectric liners disposed in the polysilicon layer and disposed on lateral sidewalls of the spacer structure, and the dielectric liners are spaced apart from each other and face toward each other.

[0123] Another aspect of the present disclosure provides a method for fabricating a semiconductor structure. The method includes receiving a substrate; forming a bit line structure on a top surface of the substrate; forming a spacer structure on the bit line structure, wherein the spacer structure includes a sacrificial layer sandwiched by a first dielectric layer and a second dielectric layer; forming a polysilicon layer over the top surface of the substrate, and forming a pair of dielectric liners in the polysilicon layer, wherein the polysilicon layer has a first surface and a second surface opposite to the first surface; removing the sacrificial layer to form a gap between the first dielectric layer and the second dielectric layer; reducing a width of the gap; and forming a seal layer to seal the gap.

[0124] In conclusion, the application discloses a manufacturing method of a semiconductor structure and a semiconductor structure thereof. The presence of a dielectric liner in the semiconductor structure prevents charge leakage from a bit line structure to a polysilicon layer, and correctness of data stored in the bit line structure can be protected.

[0125] Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without

departing from the spirit and scope of the disclosure as defined by the appended claims. For example, many of the processes discussed above can be implemented in different methodologies and replaced by other processes, or a combination thereof.
[0126] Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the present disclosure, processes, machines, manufacture, compositions of matter, means, methods or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein, may be utilized according to the present disclosure. Accordingly, the appended claims in are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods and steps.

Claims

1. A semiconductor structure, comprising: a substrate having an isolation layer; a plurality of active areas defined by the isolation layer; a plurality of source regions and drain regions disposed in the active areas; a bit line structure disposed over the substrate; a plurality of capacitor contact structures disposed next to the bit line structure; a plurality of dielectric liners disposed in the capacitor contact structures, wherein each of the dielectric liners surrounds at least a portion of the corresponding capacitor contact structures; and a plurality of landing pad layers disposed partially covering a top surface and a sidewall of the corresponding capacitor contact structure.
2. The semiconductor structure of claim 1, wherein the bit line structure comprises a bit line bottom conductive layer disposed over the substrate, a bit line middle conductive layer disposed over the bit line bottom conductive layer, and a bit line top conductive layer disposed over the bit line middle conductive layer.
3. The semiconductor structure of claim 2, wherein the bit line bottom conductive layer is formed of polycrystalline silicon, polycrystalline germanium, polycrystalline silicon germanium, titanium, tantalum, tungsten, copper, aluminum, tungsten silicide, cobalt silicide, or titanium silicide.
4. The semiconductor structure of claim 3, wherein the bit line middle conductive layer is formed of titanium nitride or tantalum nitride.
5. The semiconductor structure of claim 4, wherein the bit line top conductive layer is formed of tungsten, tantalum, titanium, copper, or aluminum.
6. The semiconductor structure of claim 1, further comprising a bit line contact disposed in the substrate.
7. The semiconductor structure of claim 6, wherein the bit line structure is disposed on the bit line contact.
8. The semiconductor structure of claim 1, further comprising a plurality of bit line capping layers disposed over the substrate, wherein each of the bit line capping layers is disposed on the corresponding bit line structure.
9. The semiconductor structure of claim 1, wherein the capacitor contact structure comprises a capacitor contact bottom conductive layer disposed in the substrate and protruding from the substrate, a capacitor contact middle conductive layer disposed on the capacitor contact bottom conductive layer, and a capacitor contact top conductive layer disposed on the capacitor contact middle conductive layer.
10. The semiconductor structure of claim 8, further comprising a plurality of bit line spacers disposed in and protruding from the substrate, wherein the bit line spacer is between the bit line structure and the capacitor contact structure, wherein a bottom surface of the bit line spacer is coplanar with a bottom surface of the bit line contact.
11. The semiconductor structure of claim 8, further comprising a plurality of bit line spacers

disposed on the substrate, wherein the bit line spacer is between the bit line structure and the capacitor contact structure, wherein a top surface of the bit line spacer is coplanar with a top surface of the bit line contact.

12. A method for fabricating a semiconductor structure, comprising: providing a substrate; forming a bit line structure on the substrate; forming a capacitor contact structure next to the bit line structure, wherein the capacitor contact structure protrudes from the substrate; recessing a top surface of the bit line structure; and forming a landing pad layer partially covering a top surface of the capacitor contact structure and partially covering a sidewall of the capacitor contact structure.

13. The method of claim 12, further comprising: forming an isolation layer in the substrate; and defining a plurality of active areas by the isolation layer.

14. The method of claim 12, wherein the formation of the bit line structure comprises: forming a bit line bottom conductive layer over the substrate; forming a bit line middle conductive layer over the bit line bottom conductive layer; forming a bit line top conductive layer over the bit line middle conductive layer; and forming a bit line capping layer over the bit line top conductive layer.

15. The method of claim 12, wherein the formation of the capacitor contact structure comprises: forming a capacitor contact bottom conductive layer protruding from the substrate; forming a capacitor contact middle conductive layer on the capacitor contact bottom conductive layer; and forming a capacitor contact top conductive layer on the capacitor contact middle conductive layer.

16. The method of claim 12, wherein the formation of the capacitor contact structure further comprises: forming a dielectric liner along a sidewall of the capacitor contact bottom conductive layer and along a sidewall of the capacitor contact middle conductive layer, wherein the dielectric liner surrounds at least a portion of the capacitor contact bottom conductive layer and a portion of the capacitor contact middle conductive layer.

17. The method of claim 12, further comprising forming a bit line contact in the substrate, wherein the bit line structure is disposed on the bit line contact.

18. The method of claim 17, further comprising forming a plurality of bit line spacers between the bit line structure and the capacitor contact structure, wherein the bit line spacers are disposed in and protrude from the substrate, wherein a bottom surface of the bit line spacer is coplanar with a bottom surface of the bit line contact.

19. The method of claim 17, further comprising forming a plurality of bit line spacers between the bit line structure and the capacitor contact structure, wherein the bit line spacers are disposed on the substrate, wherein a top surface of the bit line spacer is coplanar with a top surface of the bit line contact.